

Henry Beale

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Education

DOCTOR OF PHILOSOPHY 2022–PRESENT

Queensland Brain Institute
Advisors: Anthony Harris, Jason Mattingley
Thesis: Determining the role of neural oscillations in perception using model-based approaches

BACHELOR OF ARTS (HONOURS) 2021

The University of Queensland
Advisors: Jason Mattingley, Anthony Harris
Thesis: EEG investigation of attention and temporal learning during integrative perceptual decision-making

BACHELOR OF ARTS (EXTENDED PSYCHOLOGY MAJOR) 2019–2020

The University of Queensland

Professional

EEG LAB MANAGER 2022–2025

Queensland Brain Institute

SENIOR RESEARCH ASSISTANT 2021–2022

Supervisor: Dr. Angela Renton

RESEARCH ASSISTANT SCHOLARSHIP 2020–2021

Supervisor: Dr. Anthony M. Harris

First author works (*in preparation*)

Beale. H., Mattingley, J. B., & Harris, A. M. The impacts of visual alpha power and phase on the psychometric function for contrast detection and spatial discrimination.

Beale. H., Mattingley, J. B., & Harris, A. M. Alpha-band oscillations modulate sensory noise in efficient visual codes.

Beale. H., Livesey, E., Sewell, D., Harris, & A. M. Eye-tracking and drift diffusion modelling tell different stories about contextual cueing.

Beale. H., Harris, A. M., & Mattingley, J. The role of selective attention and learned temporal structure during integrative perceptual decision making.

Beale. H., Gastrell, T., Williams., J. G., & Harris, A. Surround modulation as a possible mechanism for rhythmic visual behaviour.

Registered reports

Harris, A. M., **Beale. H.**, & Mattingley, J. Do oscillations in brain and behaviour reflect a common underlying mechanism? A Registered Report. *Nature Human Behaviour*. Stage 1 in principle acceptance.

Contributing author publications

Harris, A. M., & **Beale. H.** (2025). Detecting behavioural oscillations with increased sensitivity: A modification of Brookshire's (2022) AR-surrogate method. *eLife*, 14:RP106141. <https://doi.org/10.7554/eLife.106141.1>

Nestor, P. J., ... **Beale. H.**, ... Gotz, J. (accepted). A pilot safety and tolerability study of scanning ultrasound as a neuromodulation therapy in Alzheimer's disease. *Brain Communications*.

Williams, J. G., Harrison, W. J., **Beale. H.**, Mattingley, J. B., & Harris, A. M. (2024). Effects of neural oscillation power and phase on discrimination performance in a visual tilt illusion. *Current Biology*, 34 (8). <https://doi.org/10.1016/j.cub.2024.03.014>

Reviewing

Journal of Neuroscience

Conference presentations

Beale. H., Mattingley, J., & Harris, A. M. (2025). Alpha oscillations in visual cortex produce anti-Bayesian perceptual biases. *Australasian Cognitive Neuroscience Society*.

Beale. H., Harris, A. M., & Mattingley, J. (2022). The role of selective attention and temporal structure during integrative perceptual decision making. *Australasian Brain and Psychological Sciences Meeting*.

Poster presentations

Beale. H., Mattingley, J. B., & Harris, A. M. (2024). Using signal detection theory to understand the role of alpha phase in visual perception. *European Conference of Visual Perception (ECVP) & European Summer School for Visual Neuroscience*.

Beale. H., Mattingley, J. B., & Harris, A. M. (2023). Spontaneous neural oscillations and aperiodic EEG signals influence the perception of weak visual stimuli. *Australasian Cognitive Neuroscience Society (ACNS)*.

Memberships

Australasian Cognitive Neuroscience Society

Community contributions and outreach

2023–present	Brisbane Experimental Psychology Student Initiative Organiser and host of monthly meetings in Brisbane
2025	Biomedical undergraduate O-week workshop Postgraduate guest speaker

Teaching

The University of Queensland undergraduate psychology
Tutoring 1st year Research Methods (Statistics) in Psychology, 1st year Introduction to Neuroscience, & 4th year Cognitive Neuroscience